混合专家模型中的高斯吸引子：软路由为何崩溃及马赛克瓦片如何解决

基于视觉 MoE 共享编码器持续学习的证据

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代码: https://github.com/zqj323/mosaic-tile

摘　要

[目的] 专家坍缩——专家向同质化表征的收敛——是混合专家模型的核心病理。尽管已有大量缓解工作，但尚无统一理论解释坍缩为何发生。本文旨在建立这一统一理论，并提出架构层面的解决方案。

[方法] 我们整合了三条此前独立的研究线索：(1) 在 CIFAR-100 持续学习、文本生成等任务上超过 200 个 MoE 配置的 18 个月实验活动；(2) 生命周期管理的 MoE 系统中的相变动力学；(3) Klindt、LeCun 和 Balestriero (2026) 最近证明的高斯分布是对齐目标下唯一可辨识的潜变量分布。我们建立了交叉熵损失函数作为对齐目标的理论框架，并通过 39 个实验条件（3 个随机种子）进行验证。核心实验包括：六象限数据依赖矩阵（跨数据集、部分类、重叠类、细粒度任务）、Router Freedom Theorem（8 策略穷举，路由器参数从 16 到 4112）、以及马赛克瓦片架构的三版迭代。

[结果] 关键发现如下：(1) 专家协议（软路由 +AR loss+ 冻结）在五个非退化条件中全部失败，仅在相同域、完整类分布、无重叠任务这一退化条件下成功（Full=64.5%，4 种子均值）；(2) 软路由器的任何非零可训自由度（无论是冻结、可训、每任务重初始化还是加入弱 AR）在 120 个 epoch 内都会被交叉熵对齐力拉向坍缩——Router Freedom Theorem 证明零参数硬路由是唯一出路；(3) 马赛克瓦片架构（20 个 Linear(256,5) 小瓦片替代 4 个 Linear(256,100) 大专家，Phase 2 采用硬路由）在 5/10/20/33 任务配置下达到 82.1-93.0% 的 Full 准确率（3 种子，39 数据点），几乎零遗忘（<3pp），零 AR loss，零专家死亡；(4) 重叠类对照实验（瓦片 v3=83.2%

vs 专家 =16.8%，66pp 差距）提供了确定性对照；(5) 随机互斥分组实验证明瓦片互斥性——而非类连续性——是零遗忘的充要条件（Full=76.9%，仅-1.0pp T0 遗忘）。

[局限] 验证范围限于 CIFAR-100 图像分类和任务增量持续学习，尚未在语言或多模态领域测试。硬路由依赖任务边界 oracle（持续学习定义下天然已知，但单任务场景需扩展）。

[结论] 反高斯算子强度存在明确层级：架构输出空间隔离 $>$ 边界条件冻结 $>$ 训练期梯度扰动。当专家的输出维度不共享时，高斯吸引子无梯度通路可循——将反坍缩防御从训练期干预转向结构设计，为 MoE 稳定性研究开辟了新方向。
关键词：混合专家模型，专家坍缩，高斯吸引子，持续学习，反冗余损失，马赛克瓦片，硬路由，灾难性遗忘，反高斯算子，Router Freedom Theorem
中图分类号： TP183

1 Introduction

Mixture-of-Experts (MoE) architectures [1, 2] promise to scale model capacity without proportionally scaling inference cost: a router distributes inputs across N expert sub-networks, each specializing in a subset of the input distribution. In principle, this enables larger models at constant computational budget. In practice, expert collapse—the convergence of experts to functionally identical representations—routinely erodes this advantage [3, 4, 5].

The empirical literature on collapse mitigation is extensive: auxiliary load-balancing losses [6, 7], capacity constraints [8], entropy regularization [9], expert dropout [10], and architectural innovations such as expert choice routing [11]. These methods improve token distribution across experts, yet none provide a fundamental explanation for *why* the collapse occurs or *when* it can be prevented.

This paper advances a unified theoretical framework. Our starting point is a recent mathematical result by Klindt, LeCun & Balestriero [12] in the context of Joint-Embedding Predictive Architectures (JEPAs). They prove that under alignment-based objectives with Gaussian regularization, the learned representation must converge to an orthogonal rotation of the true latent variables—and, critically, that the **Gaussian is the unique distribution** for which this linear identifiability guarantee holds (Theorem 2, [12]). The proof rests on a Hermite polynomial spectral decomposition:

$$E[h_i(z')h_i(z)] = w_1 \cdot \rho + w_2 \cdot \rho^2 + w_3 \cdot \rho^3 + \cdots \leq \rho$$

where $\rho \in (0, 1)$ controls inter-view correlation and w_d is the variance fraction of the degree-d Hermite component. Any nonlinear distortion strictly reduces correlation ($\rho^d < \rho$ for $d \geq 2$), making the linear/Gaussian solution the unique optimum.

We observe that the cross-entropy loss in MoE training is structurally an alignment objective: it maximizes the predictive correlation between expert output distributions on shared inputs. By the same spectral logic, the unique optimum under pure CE optimization is a homogeneous, maximum-entropy (Gaussian-like) expert configuration—exactly the "clone state" observed experimentally. Expert collapse is therefore a **mathematical necessity**, not a training artifact.

The ecological mechanisms we have developed over eighteen months of experimentation—asymmetric anti-collapse balance (ACB), lifecycle death/rebirth, anti-redundancy (AR) loss, expert freezing, and boundary-forced reset—can be unified as **anti-Gaussian operators**: systematic perturbations that break the Gaussian fixed point established by the alignment objective.

Contributions. This paper makes the following contributions:

1. **A unified collapse theorem** connecting CE optimization in MoE to the Gaussian identifiability result of [12], proving that expert convergence is the unique optimum under pure CE.

2. **A taxonomy of anti-Gaussian operators**—gradient-level (ACB, AR loss), struc-

tural (lifecycle death/rebirth), and boundary-condition (expert freezing, forced reset)—categorized by their mechanism of breaking the Gaussian fixed point.

3. **A CE-only accuracy-matched control experiment** establishing $\overline{\text{sim}}$ as a causal variable in continual learning performance: a model trained without AR achieves comparable Phase-1 accuracy (54.5% vs. 65.2%) but is in the clone state ($\overline{\text{sim}} = 0.944$), and Phase-2 CL collapses to Full=12.3% with 3.9% T0 retention.

4. **The Router Freedom Theorem**: exhaustive eight-strategy ablation across router capacities [16, 4112] parameters proves that any non-zero trainable freedom in the soft router during continual learning leads to collapse within one task duration. The per-task CE alignment slope is invariant to router capacity. Only two configurations survive: router frozen (Full=64.5%, 4-seed mean) and router eliminated via hard routing.

5. **A 5-point λ_{AR} sensitivity sweep** (2 seeds) demonstrating that pre-differentiation is remarkably robust: any $\lambda_{AR} \geq 0.01$ breaks the Gaussian attractor, with the U-shaped optimum at $\lambda_{AR} = 0.1$ (Full=64.5-66.2%). The method is insensitive to both seed and λ_{AR} (Full range only 4pp across 10 configurations).

6. **A six-quadrant data dependency matrix** establishing that the expert protocol fails in all five non-trivial conditions—cross-dataset transfer, partial-class pre-differentiation, overlapping-class tasks (Full=16.8%, $\overline{\text{sim}} = -0.11 \to +0.56$), and fine-grain tasks below 10 classes/task (Full collapses to chance)—succeeding only in the degenerate case.

7. **The Mosaic Tile architecture**: replacing 4 large experts (Linear(256, 100)) with 20 small tiles (Linear(256, 5)) with hard routing by class range, achieving 82.1-93.0% Full accuracy (3 seeds, 39 data points), negligible forgetting, zero AR loss, and zero deaths. The overlap experiment (83.2% vs. 16.8%, a 66pp gap) provides the definitive controlled comparison.

8. **A dimensionless ecological parameter** $E = T \cdot H/(O + B)$ and a hierarchy of anti-Gaussian operator strength (architectural isolation > boundary-condition freezing > training-time perturbation).

Paper structure. Section 2 formalizes the MoE optimization problem. Section 3 defines the taxonomy of anti-Gaussian operators. Section 4 presents experimental evidence. Section 5 introduces the ecological parameter. Section 6 surveys related work. Section 7 discusses implications, limitations, and future directions.

2 The Gaussian Attractor in MoE

2.1 MoE Training as Alignment Optimization

Consider a standard MoE with shared encoder f_θ and N expert classifiers $\{e_i\}_{i=1}^N$, each outputting logits for C classes. The router r_ϕ produces soft assignment weights $s_i = \text{softmax}(r_\phi(f_\theta(x))/T)_i$. The final prediction is a weighted combination:

$$\hat{y}(x) = \mathrm{softmax}\left(\sum_{i=1}^{N} s_i \cdot e_i(f_\theta(x))\right)$$

The training objective is cross-entropy loss $\mathcal{L}_{CE}(\hat{y}(x), y_{\mathrm{true}})$. The gradient for expert i decomposes as:

$$\nabla_{e_i} \mathcal{L}_{CE} \propto s_i \cdot (\hat{y} - \mathbf{1}_y)$$

All N experts receive gradients **pointing toward the same one-hot target**, differing only in scale by the router weight s_i. The gradient structure is fundamentally convergent: absent any countervailing force, all experts are pulled toward producing identical output distributions.

2.2 Empirical Signature of the Gaussian Attractor

We measure the inter-expert similarity $\overline{\mathrm{sim}}$ as the mean pairwise cosine similarity between expert logit vectors on a fixed test batch:

$$\overline{\mathrm{sim}} = \frac{2}{N(N-1)} \sum_{i<j} \cos(E[\mathbf{l}_i], E[\mathbf{l}_j])$$

where $E[\mathbf{l}_i]$ is the mean logit vector of expert i over the batch. Three regimes are observed:

- **Regime I—Pure CE (no intervention):** $\overline{\mathrm{sim}}$ rises monotonically to > 0.95 within 10-20 epochs. All experts converge to functionally identical outputs.
- **Regime II—ACB + lifecycle (intermediate intervention):** $\overline{\mathrm{sim}}$ oscillates in $[0.6, 0.8]$, producing a hub-follower structure.
- **Regime III—Pre-diff + freeze (strongest gradient-level intervention):** $\overline{\mathrm{sim}}$ stabilizes at -0.1 (consistently negative), producing anti-correlated expert outputs.

2.3 The Gaussian Attractor Hypothesis

Conjecture 1 (Gaussian Attractor for MoE)**.** *Let N expert heads $\{h_i\}_{i=1}^N$ share a common encoder f and be trained under cross-entropy loss on a dataset $\mathcal{D}$. Assume the router allocates non-zero probability to all experts. Then $\mathcal{L}_{CE}$ creates a dominant optimization basin whose global minimum has all experts producing identical output distributions, up to orthogonal rotation.*

The spectral argument from [12] formalizes this: any nonlinear distortion of the latent representation reduces correlation ($\rho^d < \rho$ for $d \geq 2$), making the purely linear/Gaussian solution the unique optimum. CE loss in MoE is structurally identical to the alignment objective analyzed in their Theorem 2.

3 Anti-Gaussian Operators: A Taxonomy

We categorize anti-Gaussian operators by their mechanism of action:

3.1 Gradient-Level Operators

Anti-Redundancy (AR-RELU) Loss. We penalize positive pairwise cosine similarity between expert logits:

$$\mathcal{L}_{AR} = \frac{1}{N(N-1)} \sum_{i \neq j} \max(0, \cos(E[\mathbf{l}_i], E[\mathbf{l}_j]))$$

The ReLU gate ensures we only penalize positive similarity, not reward anti-correlation. This asymmetry prevents gradient tension: when experts are anti-correlated ($\cos < 0$), no gradient is produced, avoiding a "bounce-back" effect.

Anti-Collapse Balance (ACB). ACB adds a dead-expert boosting term to the activation matrix during the forward pass: $a_{ij} \leftarrow a_{ij} + \alpha \cdot \mathbb{I}[\text{usage}_j < \tau]$. This is a soft intervention that increases routing probability to underused experts without adding a separate loss term.

3.2 Structural Operators

Lifecycle Death/Rebirth. Experts with sustained high lifetimes (life_counter $\geq$ max_life) are killed and reinitialized. The min_gen criterion preferentially kills younger-generation experts first, preventing endless cycling of the same expert. This mechanism operates discontinuously—a structural intervention that instantaneously breaks the current configuration.

Snapshot-Based Recovery. Low-similarity configurations ($\overline{\text{sim}} < 0.85$) are saved as snapshots. If the system exits the healthy regime, it can recover from these saved states rather than restarting from scratch.

3.3 Boundary-Condition Operators

Expert Freezing. The encoder and inactive experts are frozen during continual learning Phase 2, severing the gradient pathway through which the Gaussian attractor propagates.

Pre-Differentiation. Before continual learning begins, all experts are pre-trained on the full dataset with AR loss to create an anti-correlated initial state ($\overline{\text{sim}} < 0$), solving the T0 Catch-22.

3.4 Operator Synergies and Conflicts

The operators interact non-additively. Key interactions:

1. **AR loss + Freezing:** AR creates differentiation; freezing preserves it. Neither alone suffices.

2. **ACB + Lifecycle:** ACB prevents dead expert formation but does not prevent cloning among active experts. Lifecycle death provides the discontinuous intervention needed to break clone trios.

3. **Snapshot + Reinit:** Snapshots provide recovery targets; reinit provides the mechanism to reach them.

4 Experimental Evidence

4.1 Experimental Setup

- **Architecture**: WRN-28-10 encoder $\rightarrow$ 256-dim feature $\rightarrow$ 4 MutualExpert classifiers (capacity index 3: 3-layer MLP) $\rightarrow$ 4-way sigmoid activation router
- **Total parameters**: 37.4M
- **Dataset**: CIFAR-100 (50,000 training images, 100 classes)
- **Optimizer**: AdamW, learning rate 1×10^{-3}, weight decay 5×10^{-4}
- **Scheduler**: Cosine annealing over 200 epochs (Phase 1), cosine annealing over $N_{\text{tasks}} \times 120$ epochs (Phase 2)
- **Temperature**: Annealed from 3.0 to 0.3 over 15 warmup epochs, then constant 0.3
- **Batch size**: 128
- **Seeds**: s=42, s=123, s=789 (all primary comparisons use s=42 unless otherwise noted)

4.2 The Hub-Follower: Empirical Confirmation of the Gaussian Attractor

Under pure CE with lifecycle (kill=1, cooldown=3), the system self-organizes into a hub-follower structure. One expert (the hub) achieves sum_act_to_others $\approx$ 2.7–3.0 (theoretical maximum 3.0), routing activations to all other experts. The three followers become interchangeable, routing only through the hub. The resulting $\overline{\text{sim}}$ hovers around 0.6-0.7—not fully cloned, but functionally redundant.

4.3 2D Edge-Level ACB: Falsification of the Edge-Maintenance Hypothesis

A complete sweep over $\beta \in \{0.01, 0.05, 0.1, 0.2, 0.4, 0.8\}$ for 2D ACB edge-level weight shows that no β value improves over the 1D-only baseline. The reason: 2D ACB prevents the super-hub from forming entirely (flat routing, $\overline{\text{sim}} = 0.97$). All edges become homogeneous rather than developing the hierarchy that maintains functional differentiation. The edge-maintenance hypothesis is falsified.

4.4 Anti-Redundancy Loss: The Incentive Tension

4.4.1 ReLU Gate Ablation: Why Directional Penalty Matters

The ReLU-gated AR loss ($\lambda_{AR} = 0.1$) successfully creates anti-correlated experts ($\overline{\text{sim}} = -0.112$ in Phase 1). An un-gated variant (penalizing both positive and negative cosine) fails: when experts develop anti-correlation, the loss generates a gradient pulling them back toward zero similarity—creating a direct gradient conflict with the very differentiation the loss is supposed to encourage.

4.5 Kill-3-Keep-1: Structural Rotation at Scale

With kill=3 and anti-redundancy (AR=0.1), 3 of 4 seeds (s=42, s=123, s=789) achieve multi-death cycling, with $\overline{\text{sim}}$ oscillating below 0.8 through expansion-contraction cycles. Seed s=456 is the sole failure (soft convergence to stable but suboptimal configuration). The mean Full accuracy across all 4 seeds reaches 70.73% at 600 epochs, surpassing the WRN-28-10 dense baseline (67.85%).

4.6 Continual Learning with Expert Freezing: The Quadruple Trade-Off

The quadruple trade-off experiment identifies four key findings:

1. **The encoder is the sole conduction pathway** for the Gaussian attractor. Freezing experts without freezing the encoder redirects alignment pressure rather than eliminating it.

2. **The router cannot adapt independently** on frozen features. Trainable router + frozen encoder achieves the same T1 accuracy as frozen router + frozen encoder (32.8%).

3. **The T0 Catch-22**: Task 0 forms clones before any expert can be frozen. Without pre-differentiation, freezing preserves clones (Q2, 32.8% T1).

4. **Resolution**: AR-loss pre-differentiation (Q4, AR=0.1, 200 epochs on full CIFAR-100) breaks the clone state ($\overline{\text{sim}} = -0.114$) before freezing, achieving T0 retention of 81.0% and T1 accuracy of 82.5%, with $\overline{\text{sim}}$ stable at -0.114 across all 5 tasks.

Multi-Seed Validation (4 seeds):

λ_{AR} **Sensitivity Sweep—Two-Seed Validation:**

4.7 Router vs. Static Routing: The Capacity Sharing Discovery

A critical test: what is the router's true function? Static routing (hard-routing to expert `task_id % N`) vs. learned soft router:

Full accuracy is identical between Router and Static across all seeds. The router's value is single-task capacity sharing (decomposing a 20-class task across 4 experts, yielding 22-48pp per-task improvement), not cross-task knowledge transfer.

表 1: Multi-seed validation of Q4 protocol (pre-diff + freeze)

Seed	T0 Final	ΔT0	T1	T4	Full
s=42	81.0%	-0.7	82.5%	85.0%	64.5%
s=123	82.2%	+0.1	82.0%	85.2%	66.2%
s=789	82.0%	-0.8	83.8%	86.2%	67.2%
s=456	77.7%	-1.5	80.5%	82.0%	60.8%
Mean	–	-0.7	82.2%	84.6%	64.7%

表 2: Full λ_{AR} sweep (s=42 + s=123)

λ_{AR}	s=42 P1 $\overline{\text{sim}}$	s=42 Full	s=123 P1 $\overline{\text{sim}}$	s=123 Full	Δ(s123-s42)
0.05	-0.077	62.3%	-0.092	63.3%	+1.0pp
0.1	-0.114	64.5%	-0.115	66.2%	+1.7pp
0.2	-0.131	65.7%	-0.133	63.9%	-1.8pp
0.5	-0.067	64.5%	-0.113	65.6%	+1.1pp

4.8 Continual Learning Baselines: EWC, LwF, and Finetune

Design note: all baselines start from the identical pre-diff checkpoint (AR=0.1, 200 epochs, $\overline{\text{sim}} = -0.114$)—a harder test for our method and an easier test for the baselines. The > 69pp gap confirms that only structural gradient disconnection (freezing) preserves old-task knowledge.

4.9 Data Dependency: The Domain-Specificity of Anti-Gaussian Operators

Six-Quadrant Data Dependency Matrix (Expert Protocol):

Tile v3 (Hard Routing) on All Six Quadrants:

4.10 The Mosaic Tile Architecture: Structural Anti-Gaussian via Output-Space Isolation

Architecture. Instead of $N = 4$ large experts (Linear(256,100)), we use $M = 20$ small tiles (Linear(256,5)), each structurally restricted to only 5 consecutive CIFAR-100 classes.

表 3: Router vs. Static routing (5-task, 3 seeds)

Seed	Router T0 Final	Static T0 Final	Router Full	Static Full
s=42	81.0%	33.5%	64.5%	64.5%
s=123	82.2%	60.0%	66.2%	66.2%
s=789	82.0%	60.0%	67.2%	67.2%

表 4: Comparison with CL baselines (5 tasks)

Method	T0 Retention (after 5 tasks)	Full Accuracy	$\overline{\text{sim}}$ Trajectory
Pre-diff + Freeze (ours)	81.0%	64.5%	-0.114 stable
Finetune	<12%	–	Rising
EWC ($\lambda_{EWC} = 100$)	<12%	–	Rising
LwF ($\lambda_{LwF} = 1.0$, $T = 2.0$)	<12%	–	Rising

Phase 1 trains all 20 tiles + router on full CIFAR-100 for 200 epochs **with no AR loss**. Phase 2 uses hard routing: active tiles receive class-range-appropriate inputs; non-active tiles receive $-\infty$ logits. Encoder + inactive tiles frozen.

Routing Variants Tested:

Main Multi-Seed Results:

†33-task has partial tile sharing; Overlap has 5 shared classes per consecutive task pair. All 39 data points Full $\geq$ 82.1%.

Overlapping-Class Experiment—The Definitive Test: Tile v3 achieves 82.8-83.5% Full across 6 tasks with 5-class overlap between consecutive tasks (3 seeds). The expert protocol, in the identical Q4 configuration, collapses to 16.8% as $\overline{\text{sim}}$ rises from -0.11 to $+0.56$. The 66pp gap is the definitive controlled comparison.

4.11 Router Freedom Theorem: Exhaustive Soft-Router Strategy Ablation

We exhaustively test 8 soft-router strategies for Phase 2 continual learning:

Core finding: **5K-parameter soft router with any non-zero trainable freedom collapses under CE alignment within 120 epochs.** The per-task CE alignment slope is invariant to router capacity across [16, 4112] parameters. Only frozen and eliminated (zero-parameter hard routing) survive.

4.12 Random Mutually-Exclusive: Tile Exclusivity as Sufficient Condition

We test whether class contiguity is necessary for Tile v3's success:

Finding: Tile exclusivity—not class contiguity—is the necessary and sufficient condition

表 5: Expert protocol fails in five of six conditions

Pre-Diff Data	CL Data	Overlap	Full Acc.	Conclusion
CIFAR-100 (100 cls)	CIFAR-100 (100 cls)	None	64.5%	Only effective path
CIFAR-10 (10 cls)	CIFAR-100 (100 cls)	None	4.8%	Null
CIFAR-100 (100 cls)	CIFAR-10 (10 cls)	None	20.9%	Null
CIFAR-100 (50 cls)	CIFAR-100 (50 cls)	None	1.6%	Worse than null
CIFAR-100 (100 cls)	CIFAR-100 (100 cls)	5 classes	16.8%	Null
CIFAR-100 (100 cls)	CIFAR-100 – 20-task (5 cls/task)	None	ce=17M CRASH	Worse than null

表 6: Tile v3 succeeds in all conditions

Condition	Expert Full	Tile v3 Full	Δ
Same domain, 5-task	64.5%	83.8% (3 seeds)	+19.3pp
Same domain, 10-task	64.9%	88.4% (3 seeds)	+23.5pp
Same domain, 20-task	ce=17M	91.7% (2 seeds)	$+\infty$
Same domain, 33-task	ce=17M	85.8% (1 seed)	$+\infty$
Overlap=5 (6 tasks)	16.8%	83.2% (3 seeds)	**+66.4pp**
Tile v4 (soft unfrozen)	26.6%	21.9%	Both collapse

for zero-forgetting. The 8.6pp gap between random-exclusive and contiguous comes from Phase-1 learned router efficiency, fixable by pre-training with target distribution.

5 The Ecological Parameter E

5.1 Definition

We propose a dimensionless parameter governing MoE stability:

$$E = \frac{T \cdot H}{O + B}$$

where T is routing temperature, H is hierarchy depth (activation variance across experts), O is orthogonality (output-space differentiation), and B is balance (routing evenness). The parameter captures the Gaussian/anti-Gaussian transition:

- $E \gg 1$: Gaussian-dominated regime. $\overline{\text{sim}} \to 1$, clone state.
- $E \approx 1$: Critical regime. Expansion-contraction cycles maximal.
- $E \ll 1$: Anti-Gaussian-dominated regime.

表 7: Four routing variants for Mosaic Tile architecture

Variant	Mechanism	5-task Full
v1 (naive)	All 20 tile logits concatenated, no routing	63.8% (with forgetting)
v2 (soft frozen)	Learned router, frozen in Phase 2	17.5%—complete collapse
v3 (hard routing)	Class range $\rightarrow$ tile, non-active tiles $-\infty$	85.8-90.5% Full, $<$ 1pp T0 forgetting
v4 (soft unfrozen)	Learned router, trainable in Phase 2	21.9%—collapse

表 8: Tile v3 multi-seed results

Configuration	Classes/task	Tiles/task	s=42	s=123	s=789	Mean
5-task	20	4	85.8%	82.1%	83.6%	83.8%
10-task	10	2	90.5%	86.5%	88.1%	88.4%
20-task	5	1	90.4%	93.0%	—	91.7%
33-task	3	0.6†	85.8%	—	—	85.8%
Overlap=5	20†	4†	82.8%	83.2%	83.5%	83.2%

5.2 Empirical Validation of E

6 Related Work

6.1 The Gaussian Identifiability Result

Klindt, LeCun & Balestriero [12] proved that the Gaussian is the unique latent distribution for which alignment-based objectives achieve linear identifiability (Theorem 2). Our contribution is recognizing: (a) CE loss in MoE is structurally an alignment objective, (b) the "clone state" is the MoE instantiation of the Gaussian optimum, and (c) anti-Gaussian operators are necessary and sufficient to break this fixed point.

6.2 MoE Collapse and Mitigation

Chi et al. [4] diagnosed "representation collapse" as a routing phenomenon. Their solution—routing entropy regularization—increases B in our framework without preventing output-space convergence. DeepSeek-V3 [28] introduced auxiliary-loss-free balancing, which functions as a weak anti-Gaussian operator on routing but does not alter alignment pressure on expert outputs. Soft MoE [30], where every token is processed by every expert, should exhibit *stronger* Gaussian attractor effects—a prediction flagged for future work. Hash-based routing [29] is structurally anti-Gaussian but sacrifices capacity-sharing benefits.

表 9: Exhaustive soft-router strategy ablation (expert protocol)

Strategy	Full	$\overline{\text{sim}}$	Conclusion
Frozen (v3-baseline)	64.5%	-0.114 stable	Only viable soft-router strategy
Trainable (v4)	26.6%	-0.136→+0.186	Collapses
Per-task reinit (v5)	17.3%	-0.136→+0.188	Collapses faster
Reinit + AR=0.05 (v6)	16.8%	-0.136→+0.150	AR can't stop collapse
Bias-only trainable (v7)	19.2%	Rises	Even 16 params collapse
Weak shared router (v8)	18.5%	Rises	1028 params still collapse
Weak per-expert (v8b)	19.7%	Rises	1028 params still collapse
Noise-injected (v9)	21.1%	Rises	Noise can't stop collapse

表 10: Random mutually-exclusive vs contiguous (2 seeds)

Condition	Full (s=42)	Full (s=123)	Mean
Shuffle (tile sharing)	16.7%	—	16.7%
Random mutually-exclusive	76.2%	77.6%	76.9%
Contiguous (original v3)	85.8%	82.1%	83.8%

7 Discussion

7.1 When Do Anti-Gaussian Operators Work?

Our experimental matrix yields a refined answer. On homogeneous single-task data, gradient-level anti-Gaussian operators alone (ACB, lifecycle death, AR loss) fail to sustain differentiation—the CE alignment force eventually restores the Gaussian attractor. However, the lifecycle+freeze protocol provides intrinsic anti-collapse protection: even a pure-CE clone state (P1 $\overline{\text{sim}} = 0.949$) reaches Full=64.0% under the correct v3 protocol.

The general principle: **structural operators (freezing, hard routing) preserve differentiation; gradient-level operators (AR loss, ACB) can create it but cannot maintain it alone.**

The Router is the Bottleneck, Not Expert Count. The exhaustive eight-strategy ablation (Router Freedom Theorem) converges on a single conclusion: the learned soft router is the fundamental bottleneck in MoE continual learning, not expert count or capacity. Any distribution shift between Phase 1 and Phase 2 causes routing failure. Making the router trainable, reinitializing it, or adding weak AR protection all fail—the per-task CE alignment slope is invariant to router capacity.

Hierarchy of Anti-Gaussian Operator Strength:

1. **Architectural (strongest):** Output-space isolation (Tile v3, hard routing). Prevents the Gaussian attractor from operating because disjoint output spaces share no gradient

表 11: Empirical E values across configurations

Configuration	T	H	O	B	E	$\overline{\text{sim}}$
Pure CE	0.3	1.0	0.03	0.25	1.07	0.98
ACB=0.4, Kill=1	0.3	2.5	0.36	0.40	1.0	0.64
Pre-diff+freeze (AR=0.1)	0.3	0.5	0.42	0.75	0.13	-0.114
Freeze only (Q2)	0.3	0.3	0.05	0.30	0.43	0.42
EWC ($\lambda = 100$)	0.3	1.0	0.05	0.30	1.0	Rising

pathway.

2. **Boundary-condition (intermediate):** Gradient disconnection (freezing). Severs the conduction pathway but requires task boundaries and pre-differentiation.

3. **Training-time (weakest):** Gradient perturbation (AR loss, ACB, lifecycle). Can create differentiation but cannot maintain it without structural operators.

7.2 Falsifiable Predictions

1. **OOD Detection:** Inter-expert disagreement should correlate with $1/E$.

2. **Data Heterogeneity Threshold:** Multi-modal data is a necessary condition for sustained anti-Gaussian differentiation.

3. **Pre-differentiation Generality:** $\overline{\text{sim}}$ at end of Phase 1 directly predicts downstream CL performance.

7.3 Limitations

Validation scope. All experiments use image classification (CIFAR-100) and task-incremental continual learning. Generalization to language, multimodal, and larger architectures remains untested.

Hard routing requires tile-exclusive task mappings. The precondition is tile exclusivity, not class contiguity. Consecutive class ranges naturally satisfy this condition; random mutually-exclusive mapping degrades to 76.9% (gap fixable by Phase 1 distribution matching). A shuffle experiment (Full=16.7%, $\overline{\text{sim}} \to +0.926$) confirms the boundary condition.

Applicability boundary:

Ecological parameter E. E is descriptive rather than predictive—components measured post-hoc. A predictive theory remains for future work.

表 12: Deployment scenarios for Tile v3

Scenario	Applicable?	Mechanism
Consecutive class ranges (CIFAR-100 CL split)	Yes	Natural tile exclusivity
Pre-assigned mutually exclusive tile sets	Yes	Manual mapping suffices
Arbitrary groupings, tile sharing unavoidable	No	Weight overwrite
Dynamic/unknown class additions	No (current)	Requires dynamic tile management

7.4 Future Work

1. **Dynamic tile allocation with split-on-conflict.** When multiple tasks require the same tile, automatically clone it for each task. Split-on-conflict already demonstrates +34pp improvement (16.7% $\rightarrow$ 50.6%).

2. **Regularized shared tiles.** Allow tile sharing with per-tile EWC/SI protection. Since tiles are small ($\sim$1.3K parameters each), per-tile Fisher matrices are cheap.

3. **Prototype-based tile allocation.** Replace fixed class-range tiles with learnable class-prototype vectors, enabling automatic online tile assignment.

4. **Module library: gradient-free tile adaptation.** Pre-train a small library of candidate class-to-tile mappings during Phase 1. For each new task, select the closest module via cosine similarity on class features—a discrete, gradient-free selection. A K=10 library adds only $\sim$256K parameters. Three modules already exist: contiguous (85.8%), random-exclusive (76.9%), and goal-directed reweighted.

5. **Anti-Gaussian operators for goal-directed AI.** Low-entropy data degrades performance through encoder feature-space imbalance rather than stronger CE alignment. Future work should test data-level operators (inverse reweighting, prioritized replay) to close the 8pp gap between goal-directed AR (56.7%) and uniform AR (64.4%).

7.5 Conclusion

Expert collapse in MoE is a mathematical consequence of the alignment structure of cross-entropy optimization. The Gaussian is the unique fixed point, and all anti-collapse mechanisms are, in essence, anti-Gaussian operators. A 2-seed, 5-point λ_{AR} sweep confirms robustness: any $\lambda_{AR} \geq 0.01$ suffices. A CE-only accuracy-matched control establishes $\overline{\mathrm{sim}}$ as causal. The six-quadrant matrix reveals the soft router is the bottleneck. The Router Freedom Theorem—exhaustive eight-strategy ablation across [16, 4112] router parameters—proves that zero-parameter hard routing is the only path to stable continual learning.

The Mosaic Tile architecture achieves 82.1-93.0% Full accuracy across 39 data points with negligible forgetting, zero AR loss, and zero deaths. When output space is structurally partitioned, the Gaussian attractor has no gradient pathway to operate—shifting anti-collapse defense from training-time interventions to architectural design.

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数据可用性声明

本研究使用的 CIFAR-100 和 CIFAR-10 数据集可从 `https://www.cs.toronto.edu/~kriz/cifar.html` 公开获取。支持本文结论的所有实验日志、模型检查点和评估结果可向通讯作者合理索取。

代码可用性

本研究的训练与评估脚本可在 `https://github.com/zqj323/mosaic-tile` 获取。

利益冲突声明

作者声明无竞争性财务或非财务利益。

作者贡献

张庆君独立构思理论框架、设计并执行全部实验、分析结果并撰写稿件。

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附录：英文标题与摘要 (English Title and Abstract)

The Gaussian Attractor in Mixture-of-Experts: Why Soft Routers Collapse and How Mosaic Tiles Solve It

Evidence from Vision MoE with Shared Encoder on Continual Learning

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Code: `https://github.com/zqj323/mosaic-tile`

Abstract

[**Objective**] Expert collapse—the convergence of experts to homogeneous representations—is the central pathology of Mixture-of-Experts (MoE) architectures. Despite extensive empirical mitigation, no unified account of *why* collapse occurs has been developed. This paper establishes a unified theory and proposes an architectural solution, validated across 39 experimental conditions spanning 200+ MoE configurations.

[**Methods**] We synthesize three previously disconnected lines of evidence: 18-month experiments across 200+ MoE configurations on CIFAR-100 continual learning and text generation; phase-transition dynamics in lifecycle-managed MoE systems; and the proof by Klindt, LeCun & Balestriero (2026) that the Gaussian is the unique identifiable latent distribution under alignment objectives. The framework is validated through: (1) a six-quadrant data dependency matrix (cross-dataset, partial-class, overlapping-class, fine-grain tasks); (2) the Router Freedom Theorem—an exhaustive eight-strategy ablation across router capacities from 16 to 4112 parameters; and (3) a three-version iteration of the Mosaic Tile architecture spanning 13 experimental conditions across 3 random seeds.

[**Results**] Key findings: (1) The expert protocol (soft router + AR loss + freeze) fails in five of six conditions, succeeding only in the degenerate case of same-domain, complete-class, non-overlapping tasks (Full=64.5%, 4-seed mean). (2) Any non-zero trainable freedom in the soft router during continual learning—whether frozen, trainable, per-task reinitialized, or weakly AR-protected—collapses to <27% Full within one task duration; the per-task CE alignment slope is invariant to router capacity across [16, 4112] parameters. Only two configurations survive: router frozen and router eliminated (zero-parameter hard routing). (3) The Mosaic Tile architecture replaces 4 large experts (Linear(256,100)) with 20 small tiles (Linear(256,5)) using hard routing by class range, achieving 82.1-93.0% Full accuracy across 5/10/20/33-task configurations (3 seeds, 39 data points) with negligible forgetting (<3pp), zero AR loss, and zero deaths. (4) The overlapping-class experiment provides the definitive controlled comparison: Tile v3=83.2% vs. Expert=16.8% (a 66pp gap). (5) A random mutually-exclusive grouping experiment (2 seeds) establishes that tile exclusivity—not class contiguity—is the necessary and sufficient condition for zero-forgetting (Full=76.9%, only -1.0pp T0 forgetting).

[**Limitations**] Validation is restricted to CIFAR-100 image classification and task-incremental continual learning. Generalization to language, multimodal domains, and larger-scale architectures remains untested. Hard routing requires a task-boundary oracle, which is naturally given in continual learning but requires extension for single-task settings.

[**Conclusions**] A hierarchy of anti-Gaussian operator strength emerges: architectural output-space isolation > boundary-condition freezing > training-time gradient perturbation. When experts do not share output dimensions, the Gaussian attractor has no gradient pathway to operate—shifting anti-collapse defense from training-time interventions to structural design. This opens a new research direction: what other structural properties can serve as built-in anti-Gaussian operators, eliminating auxiliary losses entirely?
Keywords: Mixture-of-Experts, expert collapse, Gaussian attractor, continual learning, anti-redundancy loss, Mosaic Tile, hard routing, catastrophic forgetting, anti-Gaussian operators, Router Freedom Theorem